Data and Algorithm Analysis

Chapter 30 — Polynomial and the FFT

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Outline

Overview

Polynomials

Polynomial Representations

The DFT and FFT

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Polynomials

Polynomial Representations

The DFT and FFT

Overview

Two ways to represent polynomials of degree n:

1. Coefficient representation. Multiplication takes $\Theta(n^2)$ time.
2. Point-value representation. Multiplication takes $\Theta(n)$ time.

Will see how to convert back and forth in $\Theta(n \lg n)$ time by using the fast Fourier transform (FFT), which will yield a $\Theta(n \lg n)$-time algorithm for multiplication in coefficient form.

Fourier Transforms

Commonly used in signal processing.

Signal is given in the **time domain** mapping time to amplitude.

Fourier analysis expresses the signal as a weighted sum of phase-shifted sinusoids of varying frequencies.

Weights and phases associated with the frequencies characterize the signal in the **frequency domain**.

FFT is a fast way to compute a Fourier transform.

Goal: Design a circuit to compute the FFT.

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Polynomial

A polynomial in the variable $x \in F$ over an algebraic field F represents a function $A(x)$ as a formal sum

$$A(x) = \sum_{j=0}^{n-1} a_j x^j$$

where $a_0, a_1, \ldots, a_{n-1} \in F$ are polynomial coefficients.

Typically, $F = \mathbb{C}$ (complex numbers).

Polynomial A(x) has **degree** k ($\deg(A) = k$) if its highest non-zero coefficient is a_k.

Any integer strictly greater than the degree of a polynomial is a **degree-bound** of the polynomial.

Arithmetic Operations on Polynomials

Addition: If $A(x)$ and $B(x)$ are deg-n polynomials, their sum is a deg-n polynomial $C(x)$ such that $C(x) = A(x) + B(x)$ for all $x \in F$.

Let $A(x) = \sum_{j=0}^{n-1} a_j x^j$, $B(x) = \sum_{j=0}^{n-1} b_j x^j$, then

$$C(x) = \sum_{j=0}^{n-1} c_j x^j = \sum_{j=0}^{n-1} (a_j + b_j) x^j$$

Example

Let $A(x) = 6x^3 + 7x^2 - 10x + 9$, $B(x) = -2x^3 + 4x - 5$.
Then, $C(x) = A(x) + B(x) = 4x^3 + 7x^2 - 6x + 4$.

Arithmetic Operations on Polynomials

Multiplication: If $A(x)$ and $B(x)$ are deg-n polynomials, their **product** is a deg-$(2n-1)$ polynomial $C(x)$ such that $C(x) = A(x)B(x)$ for all $x \in F$.

Let $A(x) = \sum_{j=0}^{n-1} a_j x^j$, $B(x) = \sum_{j=0}^{n-1} b_j x^j$, then

$$C(x) = \sum_{j=0}^{2n-2} c_j x^j = \sum_{j=0}^{2n-2} \left[\sum_{k=0}^{j} (a_k + b_{j-k}) \right] x^j$$

Example

Let $A(x) = 6x^3 + 7x^2 - 10x + 9$, $B(x) = -2x^3 + 4x - 5$.
Then, $C(x) = A(x)B(x) = \ldots$ (on board).

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Coefficient Representation

Coefficient representation of a polynomial $A(x) = \sum_{j=0}^{n-1} a_j x^j$ is a vector of coefficients $\vec{a} = (a_0, a_1, \ldots, a_{n-1})$.

Evaluating the polynomial $A(x)$ at a given point x_0 consists of computing the value of $A(x_0)$ in Θn time using **Horner's rule**:

$$A(x) = a_0 + x_0 \left(a_1 + x_0 \left(a_2 + \cdots + x_0 \left(a_{n-1} + x_0 \left(a_{n-1} \right) \right) \ldots \right) \right)$$

Coefficient Representation

Let $\vec{a} = (a_0, a_1, \ldots, a_{n-1})$ and $\vec{b} = (b_0, b_1, \ldots, b_{n-1})$ be coefficient vectors of polynomials $A(x)$ and $B(x)$.

Polynomial addition takes $\Theta(n)$ time:

- Compute the coefficient vector $\vec{c} = (c_0, c_1, \ldots, c_n)$, where $c_j = a_j + b_j$ for $j = 0, 1, \ldots, n-1$.

Polynomial multiplication takes $\Theta(n^2)$ time:

- Compute the coefficient vector $\vec{c} = (c_0, c_1, \ldots, c_{2n-2})$, where $c_j = \sum_{k=0}^{j}(a_k b_{j-k})$ for $j = 0, 1, \ldots, 2n-2$.
- $\vec{c}$ is called the **convolution** of $\vec{a}$ and $\vec{b}$, i.e., $\vec{c} = \vec{a} \otimes \vec{b}$.

Point-Value Representation

A **point-value representation** of a deg-n polynomial $A(x)$ is a set of n *point-value pairs*

$$\{(x_0, y_0), (x_1, y_1), \ldots, (x_{n-1}, y_{n-1})\}$$

such that

- All $x_0, x_1, \ldots, x_n$ are distinct.
- $y_k = A(x_k)$ for $k = 0, 1, \ldots, n-1$.

Point-Value Representation

A polynomial has many different point-value representations.

Any set of n distinct points serves as a **basis** for the representation.

To compute a point-value representation, select n distinct points $x_0, x_1, \ldots, x_{n-1}$ and then evaluate $A(x_k)$ for $k = 0, 1, \ldots, n-1$.

- With Horner's method, evaluating at n points takes $\Theta(n^2)$ time.
- By choosing appropriate x_k, this can run in $\Theta(n \lg n)$ time.

Determining the coefficient form of a polynomial from a point-value representation is called **interpolation**.

Interpolation is well defined when the polynomial has a degree-bound *equal* to the given number of point-value pairs.

Point-Value Representation

Theorem (Uniqueness of an interpolating polynomial)

For any set $\{(x_0, y_0), (x_1, y_1), \ldots, (x_{n-1}, y_{n-1})\}$ *of n point-value pairs such that all the* x_k *values are distinct, there is a unique deg-n polynomial* $A(x)$ *such that* $y_k = A(x_k)$ *for* $k = 0, 1, \ldots, n-1$.

Proof.

On board □

Point-Value Representation

Let $\{(x_0, y_0), (x_1, y_1), \ldots, (x_{n-1}, y_{n-1})\}$ and $\{(x_0, y'_0), (x_1, y'_1), \ldots, (x_{n-1}, y'_{n-1})\}$ be the point-value representations for $A(x)$ and $B(x)$.

Addition: A point-value representation for $C(x) = A(x) + B(x)$ is

$$\{(x_0, y_0 + y'_0), (x_1, y_1 + y'_1), \ldots, (x_{n-1}, y_{n-1} + y'_{n-1})\}$$

Addition of two polynomials in point-value representation takes $\Theta(n)$ time.

Point-Value Representation

Multiplication: Given $A(x), B(x)$, and their point-value representation, compute $C(x) = A(x)B(x)$.

Observation: If $C(x) = A(x)B(x)$, $C(x_k) = A(x_k)B(x_k)$ for any x_k.

- Point-wise multiply point-value representations of A and B to obtain point-value presentation for C.

Recall: In polynomial multiplication, $\deg(C) = \deg(A) + \deg(B)$.

- If A and B have degree-bound n, C has degree-bound $2n$.

Point-value representation for A and B consists of n point-value pairs
$\Rightarrow$ Point-wise multiplication only gives n point-value pairs.

- Need $2n$ pairs to interpolate C. ***How?***

Point-Value Representation

Extend point-value representations for A and B consisting of $2n$ point-value pairs each.

Let $\{(x_0, y_0), (x_1, y_1), \ldots, (x_{2n-1}, y_{2n-1})\}$ and $\{(x_0, y'_0), (x_1, y'_1), \ldots, (x_{2n-1}, y'_{2n-1})\}$ be the extended point-value representations for A and B. The a point-value representation for C is

$$\{(x_0, y_0 y'_0), (x_1, y_1 y'_1), \ldots, (x_{2n-1}, y_{2n-1} y'_{2n-1})\}$$

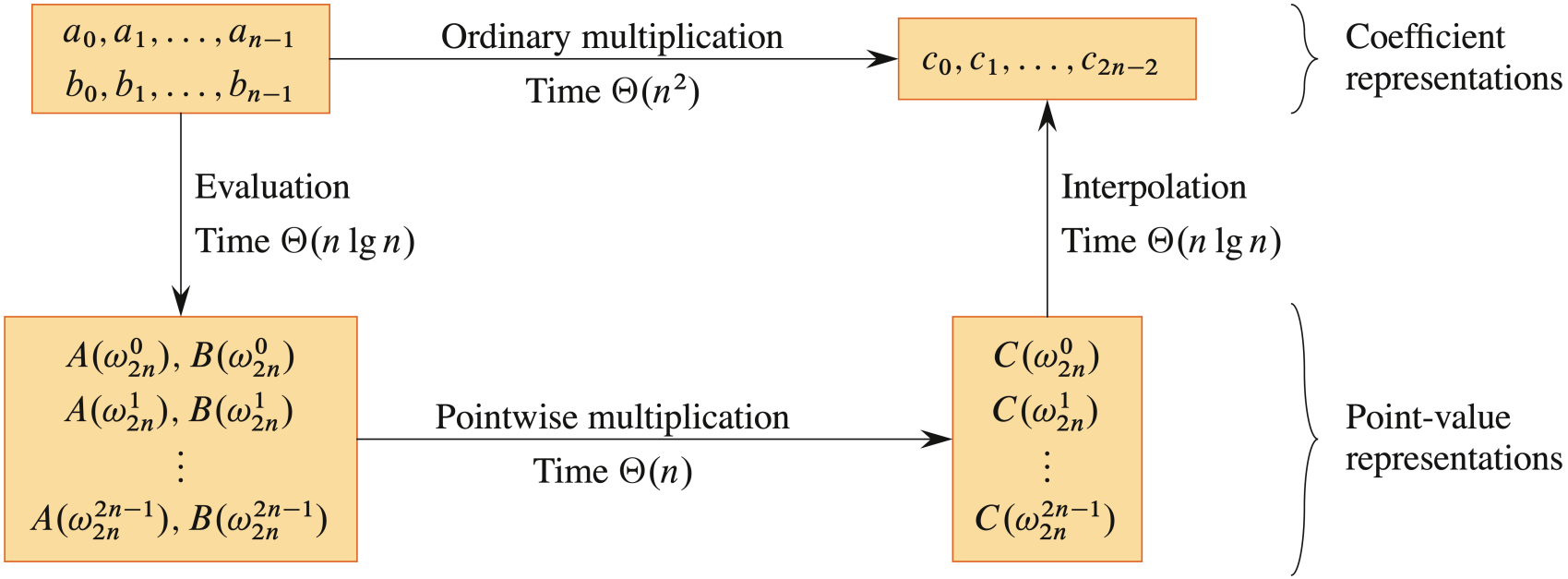
Multiplying polynomials in point-value form takes just $\Theta(n)$ time.

Question: How to extend point-value representations from n to $2n$ point-value pairs?

Fast Polynomial Multiplication in Coefficient Form

- Exploit linear-time multiplication in *point-value* form to expedite polynomial multiplication in *coefficient* form.
- Need to evaluate and interpolate quickly.
- Any points can be evaluation points $\Rightarrow$ choose special points that permit *fast* representation conversion (e.g., in $\Theta(n \lg n)$ time).
- **Complex roots of unity** as the evaluation points. Use FFT to evaluate and interpolate.

Fast Polynomial Multiplication in Coefficient Form



Fast Polynomial Multiplication in Coefficient Form

Four-step process:

1. **Double degree-bound:** Create coefficient representations of $A(x)$ and $B(x)$ as deg-$2n$ polynomials by adding n high-order zero coefficients to each. Time: $\Theta(n)$.
2. **Evaluate**: Compute point-value representations of $A(x)$ and $B(x)$ of length $2n$ with FFT of order $2n$ on each polynomial. These representations contain the values of the two polynomials at the $(2n)$th roots of unity. Time: $\Theta(n \lg n)$.

Fast Polynomial Multiplication in Coefficient Form

Four-step process:

3. **Point-wise multiply**: Compute a point-value representation for the polynomial $C(x) = A(x)B(x)$ by multiplying these values together point-wise. This representation contains the value of $C(x)$ at each $(2n)$th root of unity. Time: $\Theta(n)$.
4. **Interpolate**: Create the coefficient representation of the polynomial C(x) with FFT on $2n$ point-value pairs to compute the **inverse DFT**. Time: $\Theta(n \lg n)$.

Total time: ?

Theorem

It is possible to multiply two deg-n polynomials in $\Theta(n \lg n)$ time with both the input and output representations in coefficient form.

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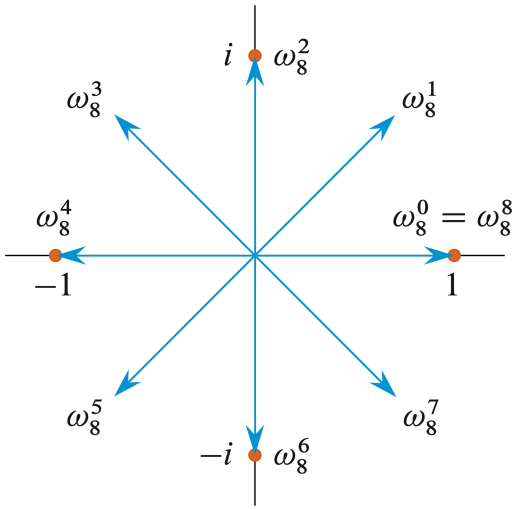
Complex Roots of Unity

A complex nth root of unity is a complex number ω such that $\omega^n = 1$.

- There are exactly n complex nth roots of unity: $e^{2\pi ik/n}$ for $k = 0, 1, \ldots, n-1$
- The **principal nth root of unity** is $\omega_n = e^{2\pi i/n}$.
- All other complex nth roots of unity are powers of ω_n.

Example



Complex Roots of Unity

The n complex nth roots of unity, $\omega_n^0, \omega_n^1, \ldots, \omega_n^{n-1}$ form a group under multiplication. This group has the same structure as the additive group $(\mathbb{Z}_n, +)$ modulo n

- $\omega_n^n = \omega_n^0 = 1$ implies that $\omega_n^j \omega_n^k = \omega_n^{(j+k) \mod n}$.
- Similarly, $\omega_n^{-1} = \omega_n^{n-1}$.

Lemma (Cancellation Lemma)

For any integers $n \geq 0, k \geq 0$, and $d > 0$, $\omega_{dn}^{dk} = \omega_n^k$.

Corollary

For any even integer $n > 0$, $\omega_n^{n/2} = \omega_2 = -1$.

Complex Roots of Unity

Lemma (Halving Lemma)

If $n > 0$ is even, then the squares of the n complex nth roots of unity are the $(n/2)$ complex $(n/2)$th roots of unity.

Lemma (Summation Lemma)

For any integer $n \geq 1$ and nonzero integer k not divisible by n, $\sum_{j=0}^{n-1} (\omega_n^k)^j = 0$.

The DFT

DFT($\vec{a}$)

Input: coefficient vector $\vec{a} = (a_0, a_1, \ldots, a_{n-1})$ of a polynomial $A(x) = \sum_{j=0}^{n-1} a_j x^j$

Output: $y_1, \ldots, y_2, \ldots, y_{n-1}$, where $y_k = A(\omega_n^k) = \sum_{j=0}^{n-1} a_j \omega_n^{kj}$ (Evaluations of A at n complex nth root of unity)

The FFT

Fast Fourier transform (FFT) uses properties of the complex roots of unity to compute $\mathrm{DFT}_n(\vec{a})$ in $\Theta(n \lg n)$ time.

Divide-and conquer strategy: Separate the <u>even</u>-indexed and <u>odd</u>-indexed coefficients of $A(x)$ to define the two deg-$(n/2)$ new polynomials $A^{\text{even}}(x)$ and $A^{\text{odd}}(x)$ of degree-bound $n/2$:

- $A^{\text{even}}(x) = a_0 + a_2 x + a_4 x^2 + \cdots + a_{n-2} x^{n/2-1}$.
- $A^{\text{odd}}(x) = a_1 + a_3 x + a_5 x^2 + \cdots + a_{n-1} x^{n/2-1}$.

$$A(x) = A^{\text{even}}(x^2) + xA^{\text{odd}}(x^2)$$

The FFT

Evaluating $A(x)$ at $\omega_n^0, \omega_n^1, \ldots, \omega_n^{n-1}$ becomes

- Evaluate the deg-$(n/2)$ polynomials $A^{\text{even}}(x^2)$ and $A^{\text{odd}}(x^2)$ at points $(\omega_n^0)^2, (\omega_n^1)^2, \ldots, (\omega_n^{n-1})^2$.
- Combine the results as $A(x) = A^{\text{even}}(x^2) + xA^{\text{odd}}(x^2)$.

FFT Procedure

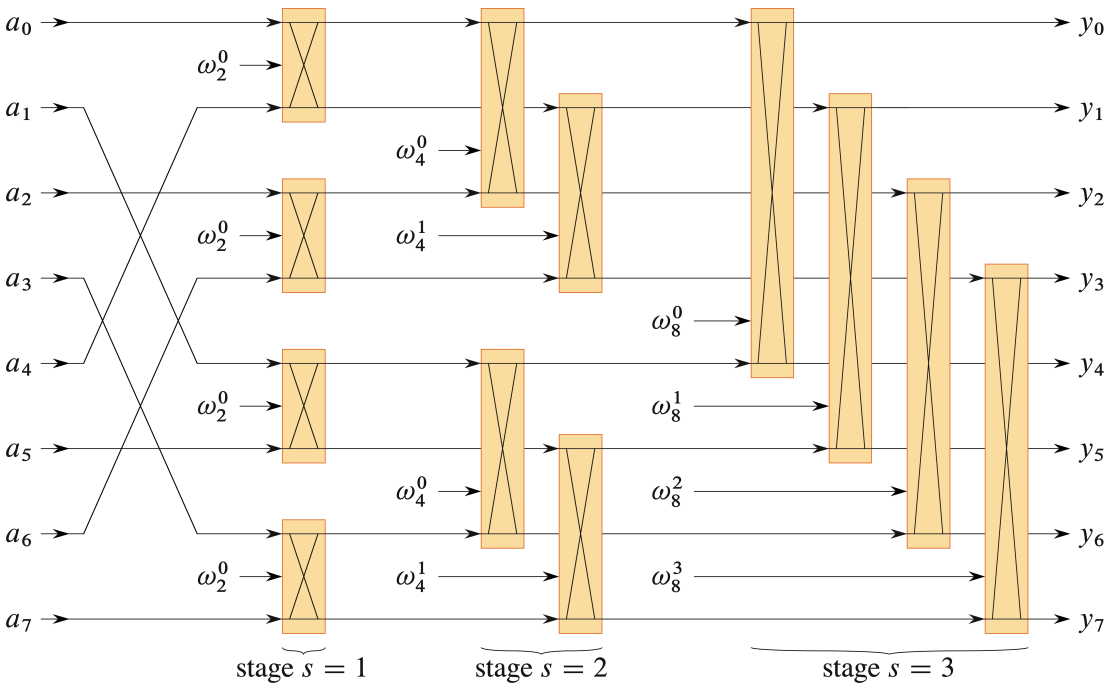
By Halving Lemma, the list $(\omega_n^0)^2, (\omega_n^1)^2, \ldots, (\omega_n^{n-1})^2$ consists **not** of n distinct values but only of $(n/2)$ complex $(n/2)$th roots of unity, with each root occurring exactly twice.

FFT recursively evaluates deg-$(n/2)$ A^{even} and A^{odd} at $n/2$ complex $(n/2)$th roots of unity.

- Subproblems have the same form as the original problem, but are half the size.
- Divide an n-element DFT_n computation into two $(n/2)$-element $\text{DFT}_{n/2}$ computations.

FFT Pseudocode

```
if n = 1
    return a
ω_n = e^{2πi/n}
ω = 1
a^even = (a_0, a_2, ..., a_{n-2})
a^odd = (a_1, a_3, ..., a_{n-1})
y^even = FFT(a^even, n/2)
y^odd = FFT(a^odd, n/2)
for k = 0 to n/2 − 1
    y_k = y_k^even + ω y_k^odd
    y_{k+(n/2)} = y_k^even − ω y_k^odd
    ω = ω ω_n
return y
```



Running Time: ?

Interpolation at the Complex Roots of Unity

For polynomial multiplication, need to convert from coefficient form to point-value form.

Evaluating the polynomial at the complex roots of unity $\Rightarrow$ pointwise multiply $\Rightarrow$ convert from point-value back to coefficient by interpolating.

How to interpolate?

Interpolation at the Complex Roots of Unity

Write the DFT as a matrix equation $\vec{y} = V_n \vec{a}$, where V_n is a Vandermonde matrix containing the appropriate powers of ω_n

$$\begin{pmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \cdots & 1 & \\ 1 & \omega_n & \omega_n^2 & \omega_n^3 & \cdots & \omega_n^{n-1} \\ 1 & \omega_n^2 & \omega_n^4 & \omega_n^6 & \cdots & \omega_n^{2(n-1)} \\ 1 & \omega_n^3 & \omega_n^6 & \omega_n^9 & \cdots & \omega_n^{3(n-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega_n^{n-1} & \omega_n^{2(n-1)} & \omega_n^{3(n-1)} & \cdots & \omega_n^{(n-1)(n-1)} \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_{n-1} \end{pmatrix}$$

For inversion, $\vec{a} = \mathrm{DFT}_n^{-1}(\vec{y})$, computes $\vec{a} = V_n^{-1}\vec{y}$.

Interpolation at the Complex Roots of Unity

Theorem

For $j, k = 0, 1, \ldots, n-1$, the (j, k) entry of V_n^{-1} is $\omega_n^{-jk/n}$.

Proof.

Just need to show that $V_n^{-1} V_n = I$ ($n \times n$ identity matrix). □

FFT Circuits

Because FFT's applications in signal processing require fast speed, often implemented as a circuit in hardware.

FFT's divide-and-conquer structure allows for a circuit with a parallel structure.

The circuit **depth** (maximum number of computational elements between an output and any input that can reach it) is $\Theta(n)$.

FFT Butterfly Operations

Recall the following lines in the **for** loop of FFT procedure:

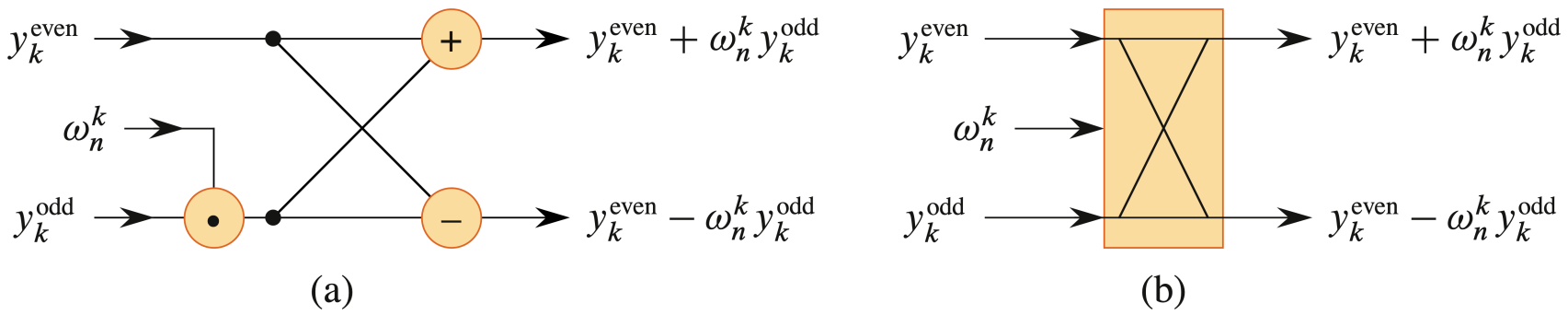
```
for k = 0 to n/2 − 1
    y_k = y_k^even + ω y_k^odd
    y_{k+(n/2)} = y_k^even − ω y_k^odd
    ω = ω ω_n
return y
```

ωy_k^{odd} is computed twice per iterationx.
Optimize the loop by pulluing out **common subexpression**.

$$t = \omega y_k^{\text{odd}}$$
$$y_k = y_k^{\text{even}} + t$$
$$y_{k+(n/2)} = y_k^{\text{even}} - t$$

This is called **butterfly operation**.

FFT Butterfly Operations



FFT Recursive Circuit Structure

Recall that FFT procedure follows divide-and-conquer strategy:

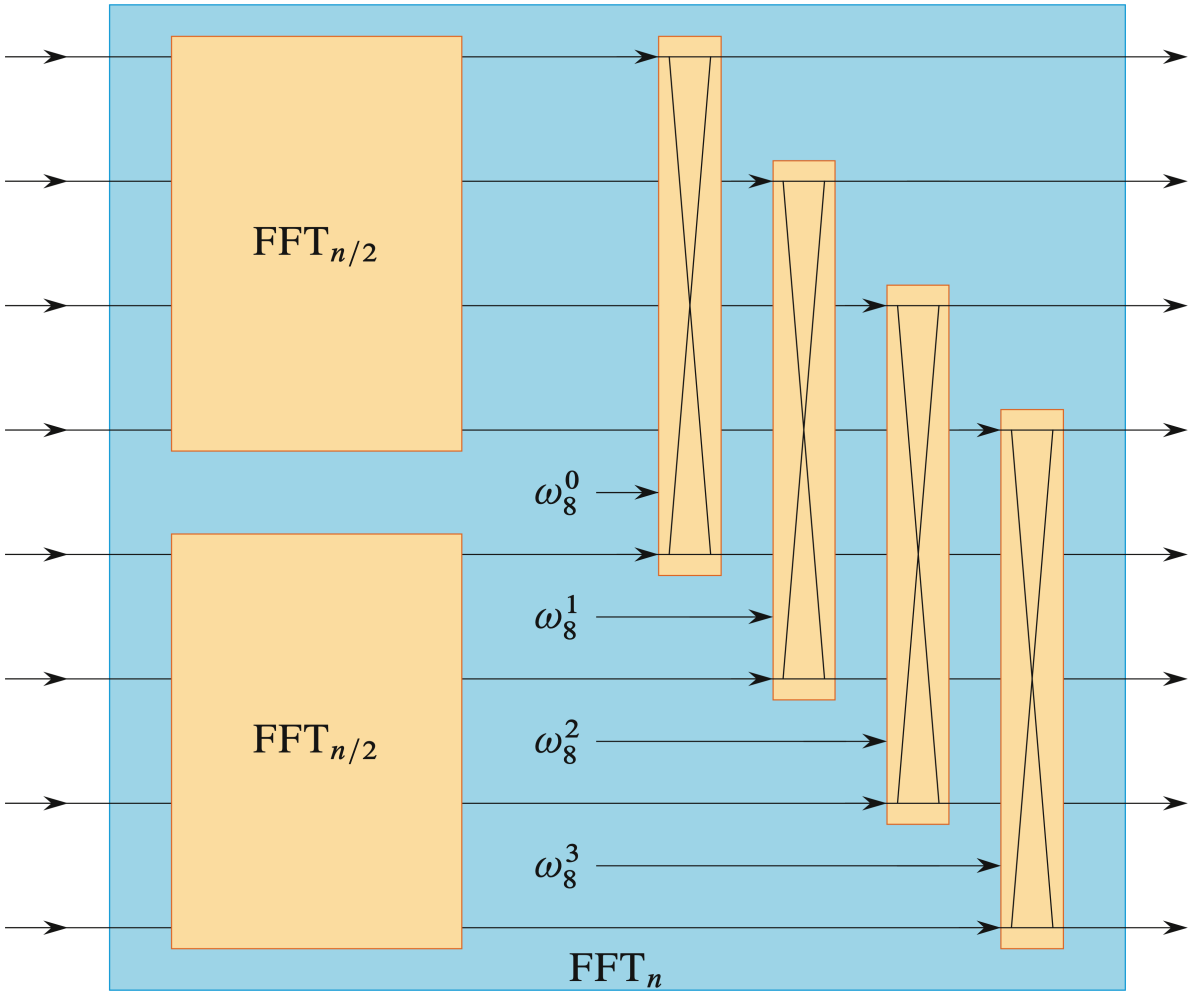
Divide the n-element input vector into its $n/2$ even-indexed and $n/2$ odd-indexed elements.

Conquer by recursively computing the DFTs of the two subproblems, each of size $n/2$.

Combine by performing $n/2$ butterfly operations.

- These butterfly operations work with twiddle factors $\omega_n^0, \omega_n^1, \ldots, \omega_n^{n/2-1}$.

FFT Recursive Circuit Structure



Permuting the Inputs

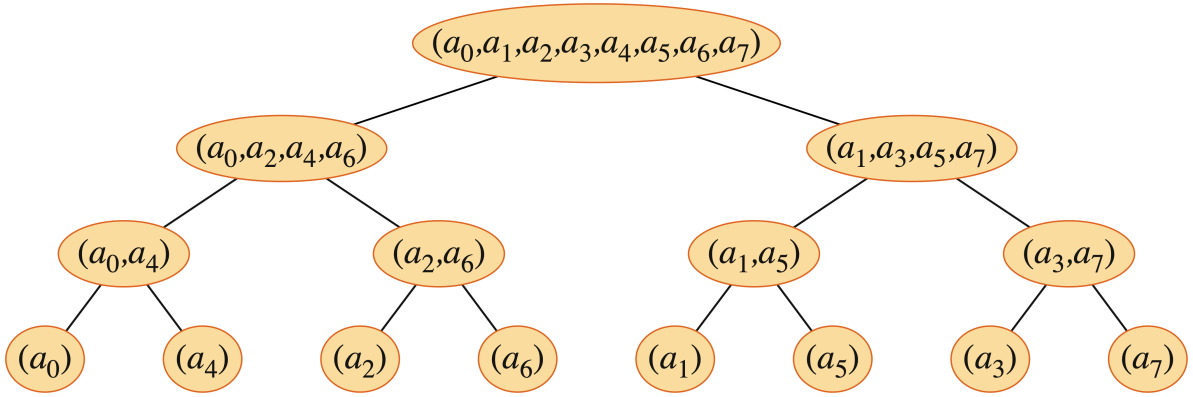
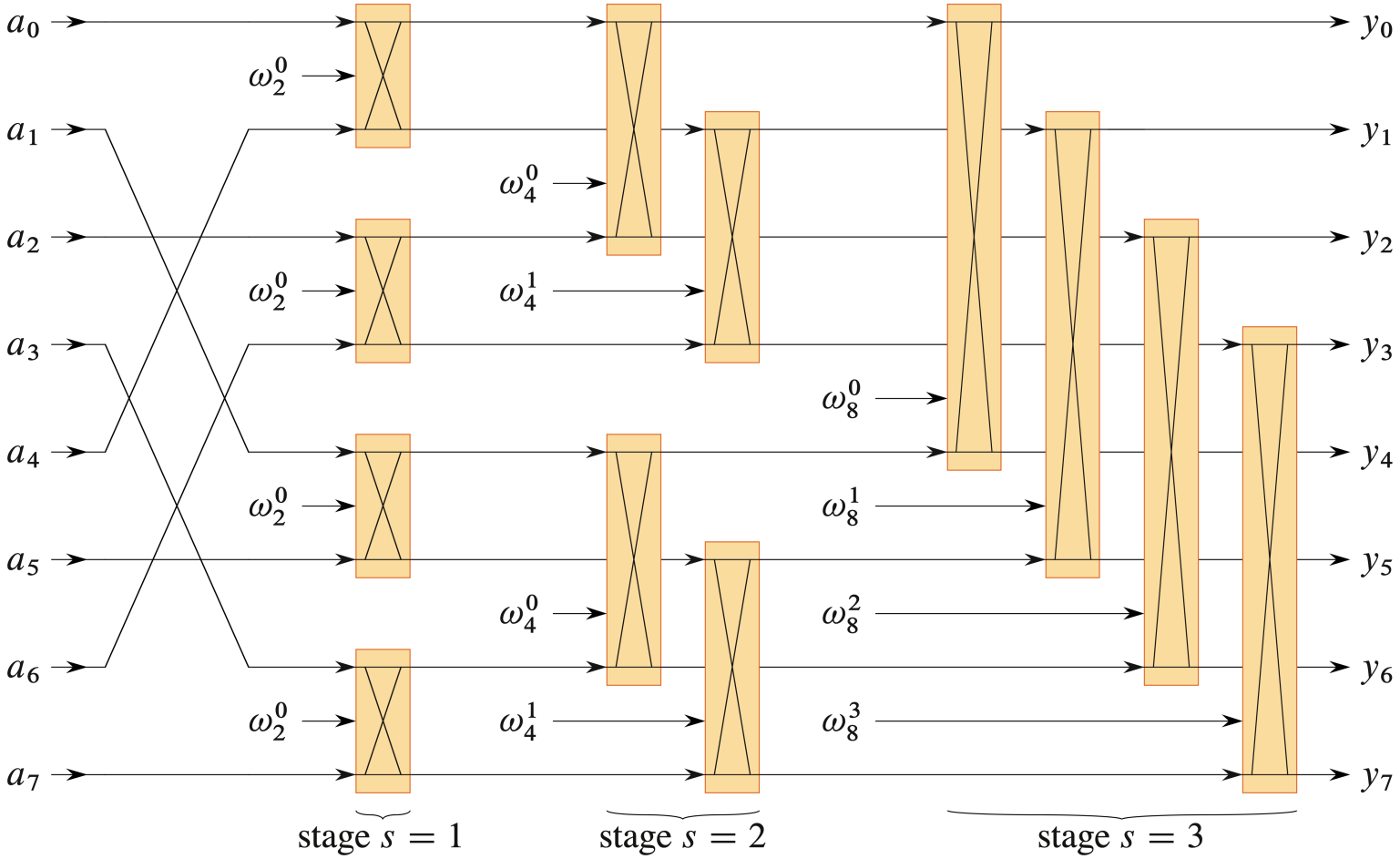
How does the circuit incorporate the divide step?

Idea: Understand how input vectors to the various recursive calls of the FFT relate to the original input vector to emulate the divide step at the start for all recursion levels.

- Bit-reversal permutation.

Example

Tree of input vectors to the recursive calls of FFT for $n = 8$:



The Full FFT Circuit